QA AUDIT & BUG REPORT

# Forma — Workout App QA Audit & Bug Report

Workout template library, custom workout builder, workout UX refresh, and an application-wide smoke test

| Prepared | Build / branch | Environment |
|---|---|---|
| 30 October 2026 | frontend: feat/workout-system · backend: feat/workout-supersets (uncommitted working tree at time of testing) | Local Vite + Express API, disposable Render Postgres (test DB), Chromium via Playwright |

## 1. Executive summary

**Release readiness:** Not yet release-ready for production. The workout experience itself is ready for a staging release; the blockers are app-wide (missing KaiMemory migration breaks coach chat, dependency advisories, text-contrast token) and environmental (no deployed API).

The workout area is in good shape after this pass. Every workout journey in scope was automated end to end against the real API and a test database, and passes: library browse / search / filter, plan and workout details, save a copy, customize, the builder (create, replace, reorder, duplicate, remove, validation, unsaved-changes protection, edit, delete), and start → log → finish → history. Thirteen workout-scope defects were found and fixed, four of them High (lost builder saves, warm-up flag on every set, wrong exercise variant saved, open workout silently discarded). What remains open is outside the workout scope and pre-existing: coach chat fails on a migrated database (missing KaiMemory migration), a text-contrast token below WCAG AA, dependency advisories, and accessibility issues on the body-map screen.

### Areas tested

- Workout template library: browse, search, filter (goal, level, equipment, duration, days/week), plan and workout details, save a copy, customize
- Custom workout builder: create, add / replace / reorder / duplicate / remove, validation, unsaved-changes protection, save, edit, delete, persistence after reload
- Workout session: start from a template, log, finish, summary, history; starting while a workout is open
- Application-wide smoke: 21 routes for runtime / console / 5xx errors, session persistence, CSV and JSON data export
- Accessibility: axe-core WCAG 2.2 A/AA scans of workout screens plus an app-wide report-only scan; keyboard-accessible names checked in tests
- Responsive: phone-width (Pixel 7 emulation) overflow checks on 7 main routes plus the builder

### Headline results

| Check | Result |
|---|---|
| Final full Playwright run (all specs, both projects) | 192 passed, 0 failed, 5 skipped (QA screenshot specs, opt-in) — 8.7 min |
| Earlier full run in this pass | 194 passed, 2 failed — both were test-side mistakes, fixed and re-run green |
| New workout specs (library 10, builder 7, session 2) | All passing in the final run |
| App-wide smoke (21 routes, reload, export, 7 phone layouts) | All passing in the final run |
| axe on workout screens (templates, builder, plan dialog, finish screen) | 0 serious / critical, colour contrast excluded (logged as QA-01) |
| App-wide axe scan, 10 routes (report only) | Contrast on 7 routes; /body has 2 further serious rules (QA-09); 3 routes clean |
| TypeScript build (tsc -b + vite build) | Passes |
| ESLint, frontend | 0 errors, 57 warnings (React Compiler readiness) |
| Backend typecheck, lint, Vitest | Pass; Vitest 119 / 119 |

### Major findings

- Before this pass the builder could not save correctly: it always submitted the blank draft from its first render (fixed).
- "Add a warm-up set" marked every set as a warm-up, so the whole exercise disappeared from volume and PRs (fixed).

- With the backend, saved workouts lost RPE / warm-up settings and six common exercises (e.g. Barbell Deadlift) were stored as a different variant (fixed).
- Starting a workout while another was open silently discarded the open one (fixed: now asks).
- App-wide (pre-existing, out of scope): coach chat returns 500 on any database built from migrations — the KaiMemory table has no migration, and the chat test accepts 500 so it never caught this (open).
- App-wide: the light theme's tertiary text colour measures 3.45:1 against the page background (WCAG AA needs 4.5:1) — 200+ elements per screen (open).
- App-wide: 18 backend dependency advisories including a critical one on the request path (proxy-addr) and 6 frontend advisories (open).

## Critical risks

- A real deploy runs `prisma migrate deploy`, so coach chat would fail for every user until the KaiMemory migration exists (QA-10).
- Dependency advisories (critical proxy-addr via Express) — must be upgraded and re-tested before any public deployment.
- No deployed API: production (GitHub Pages) runs the no-backend build; API-mode features are only verified locally against a disposable database that expires after 30 days.
- Exercise images are hot-linked from a third-party host (exercise-dataset.com); an outage or policy change blanks every thumbnail.
- Custom exercises cannot be stored in API mode (no custom-exercise endpoint); the builder now warns instead of silently dropping them.

# 2. Test coverage

## Execution environment

| Item | Detail |
|---|---|
| Frontend | Vite 5 dev server (React 18), API mode (VITE_API_URL set), port 5180 |
| Backend | Express + Prisma (tsx), port 4100, NODE_ENV=development, ANTHROPIC_API_KEY cleared for tests |
| Database | Disposable Render Postgres 16 supplied by the project owner; 13 migrations applied; seed + RepDB import (702 exercises) |
| Account | Seed account alex@forma.app (from backend/prisma/seed.ts); purchase-race tests register and delete their own users |
| Browsers | Playwright 1.63 Chromium: Desktop Chrome profile and Pixel 7 (412 px) profile |
| Automated checks | TypeScript (tsc -b, tsc --noEmit), ESLint 10, Vitest (backend), Playwright E2E, axe-core 4.14 |
| Date | 30 October 2026 |

## Result legend

**Passed** — executed and met expectations. **Failed** — executed and did not. **Blocked** — could not be executed (reason given). **Not Tested** — out of reach this pass. Static code reading is never counted as Passed.

## Journeys and checks

| Area | Journey / check | Method | Result | Notes |
|---|---|---|---|---|
| Library | Browse plans and single workouts with a live count | E2E (Playwright) | **Passed** | |
| Library | Search by workout or exercise name; clear search | E2E (Playwright) | **Passed** | |
| Library | Filter by goal, equipment and days / week; combined; empty state offers 'clear filters' | E2E (Playwright) | **Passed** | |
| Library | Filter by level and session length | — | **Not Tested** | Controls present; no automated assertion this pass |
| Library | Plan details: schedule, progression, every day; open a day; Escape returns | E2E (Playwright) | **Passed** | |
| Library | Workout details: targets, rest, timed and per-side work, frequency | E2E (Playwright) | **Passed** | |
| Library | Save a copy; original unchanged; copy survives reload | E2E (Playwright) | **Passed** | |
| Library | Customize opens the builder on a copy | E2E (Playwright) | **Passed** | |
| Library | Save all days of a plan to my workouts | E2E (Playwright) | **Passed** | |
| Library | Duplicate / rename one of my own workouts from its menu | — | **Not Tested** | Menu actions not automated |
| Media | Thumbnails play start ↔ peak frames on hover; stay still with reduced motion | E2E (Playwright) | **Passed** | |
| Builder | Create from scratch: validation, smart defaults, reorder, edit, save, persists after reload | E2E (Playwright) | **Passed** | |
| Builder | Replace with a suggested swap, duplicate, remove with undo, discard | E2E (Playwright) | **Passed** | |
| Builder | Leaving with unsaved changes asks first; kept draft is restored; discard clears it | E2E (Playwright) | **Passed** | |
| Builder | Edit and delete a saved workout | E2E (Playwright) | **Passed** | |
| Builder | Warm-up option adds exactly one warm-up set first | E2E (Playwright) | **Passed** | |
| Builder | Phone width: fits, footer not covered | E2E (Playwright) @mobile | **Passed** | |
| Builder | Custom exercise in API mode warns before saving | — | **Not Tested** | Dialog implemented; not automated (QA-03) |
| Session | Start a template, log a set, finish, summary (week, muscles, share button), history | E2E (Playwright) | **Passed** | |
| Session | Starting while a workout is open asks before replacing it | E2E (Playwright) | **Passed** | |

| Area | Journey / check | Method | Result | Notes |
|---|---|---|---|---|
| Session | Share this workout (system share sheet / clipboard) | — | Not Tested | Button presence asserted; share itself not exercised |
| Smoke | 21 routes render with no runtime, console or 5xx errors | E2E (Playwright) | Passed | |
| Smoke | Session survives a full reload | E2E (Playwright) | Passed | |
| Smoke | CSV and JSON data export (API and account page download) | E2E (Playwright) | Passed | |
| Smoke | No horizontal overflow on 7 routes at phone width | E2E (Playwright) @mobile | Passed | |
| Smoke | Existing suites: auth, dashboard, exercise library, goals & store, navigation, nutrition, progress, settings, trainer, workouts | E2E (Playwright) | Passed | The trainer 'POST /chat' test passes only because it accepts 500 (T-00) |
| A11y | axe on templates, builder, plan dialog and finish screen (serious / critical) | axe-core in E2E | Passed | color-contrast excluded and logged as QA-01 |
| A11y | App-wide axe scan, 10 routes | axe-core (report only) | Failed | QA-01 on 7 routes; QA-09 on /body |
| A11y | Screen reader pass (VoiceOver / NVDA) | — | Not Tested | |
| Coach | Send a chat message (canned reply, no API key) | E2E server log | Failed | 500 — KaiMemory table missing (QA-10); out of scope, not fixed |
| Coach | Live Kai (Claude) replies | — | Blocked | API key cleared so tests never bill a real key |
| Integrations | Payments, wearables OAuth, email delivery | — | Blocked | No provider accounts or SMTP configured |
| Platforms | Safari / WebKit, Firefox, real devices | — | Not Tested | |
| Platforms | No-backend (GitHub Pages) build end to end | — | Not Tested | |

Totals: **Passed** 23 · **Failed** 2 · **Blocked** 2 · **Not Tested** 7

# 3. Detailed bug register

Every reproducible issue found in this pass. Issues fixed during the pass because they sat directly in the workout builder / template scope are listed in section 3b; everything else is left open for the next iteration.

| ID | Title | Sev. | Pri. | Module | Reproduction steps | Expected | Actual | Evidence | Suggested fix | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| QA-10 | Coach chat fails (500) on any database built from migrations | High | P1 | Kai chat API (backend) | 1. Create a database with `prisma migrate deploy` (as the E2E stack and a real deploy do). 2. Sign in. 3. POST /chat (or send a message on /trainer). | 201 with the user message and Kai's reply (canned when no API key). | 500. recallMemories() queries KaiMemory, which is in schema.prisma but created by none of the 13 migrations (Prisma P2021 'table does not exist'). | E2E server log during trainer.spec 'POST /chat' (the test passes because it accepts 500); grep of prisma/migrations finds no KaiMemory | Add the missing migration (additive; needs approval), and make memory recall fail soft (catch → no memories) so chat never depends on it. | Open |
| QA-01 | Tertiary text fails WCAG AA contrast (light theme) | Medium | P1 | Design system (app-wide) | 1. Open any screen in the light theme. 2. Inspect any secondary label (e.g. 'my workouts', tab labels). 3. Run axe color-contrast. | Small text ≥ 4.5:1 against its background (WCAG 1.4.3). | #8C8391 on #FEF7FA = 3.45:1. axe reports 200+ nodes per screen. | smoke.spec a11y run (color-contrast: 202–215 nodes); measured in browser | Change the light theme --text-tertiary in lib/themes.ts from #8C8391 to #776E7C (4.62:1), then re-run axe. | Open |
| QA-02 | Dependency advisories incl. critical proxy-addr | High | P1 | Backend / frontend dependencies | Run `npm audit` in backend/ and frontend/. | No high or critical advisories on runtime packages. | Backend 18 (3 critical; proxy-addr on the request path). Frontend 6 (vite high, react-router-dom moderate). | npm audit output ( docs/AUDIT_REPORT.md F-17) | Move Express and deps, nodemailer, zod, prisma, vitest, vite, react-router to patched versions; re-run all suites. | Open |
| QA-03 | Custom exercises can't be stored with the backend | Medium | P2 | Builder / library API | 1. API mode. 2. Builder → add exercise → type a name not in the library → 'use as custom exercise'. 3. Save. | The custom exercise is saved with the workout. | The server has no custom-exercise endpoint; the builder now warns and offers to save without it (previously dropped silently). | lib/templates.ts unmatchedExercises; builder dialog 'some exercises can't be saved' | Add user-owned exercises (POST /library/exercises, Exercise.ownerId) and resolve to them. | Open |
| QA-04 | Exercise images hot-linked from a third-party host | Medium | P2 | Exercise media | Block exercise-dataset.com (or go offline) and open Workouts → Templates. | Thumbnails still render (bundled or self-hosted). | Every thumbnail falls back to the dumbbell glyph; images depend on an external host. | lib/repdb.ts REPDB_IMAGE_BASE | Self-host the used frames (≈ 1,000 WebP, licensed via RepDB) or bundle them with the build. | Open |
| QA-05 | Signed-out login page polls authenticated endpoints | Low | P3 | Auth / providers | Open /login signed out; watch the network panel. | No calls to authenticated endpoints before sign-in. | /store/wallet, /customization, /me/settings return 401 repeatedly (≈ 20 calls). | Browser network log (F-18) | Gate wallet / customization / settings hydration on an authenticated session. | Open |

| ID | Title | Sev. | Pri. | Module | Reproduction steps | Expected | Actual | Evidence | Suggested fix | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| QA-06 | useAction runs the first render's callback | Low | P2 | Data layer (api/hooks.ts) | Any useAction callback that reads component state instead of taking arguments. | The action uses current state. | run() is memoised with [] deps, so it calls the closure from the first render. Workouts 'generate' reads the first render's plan and session length. | api/hooks.ts useAction; builder save bug FX-01 had this root cause | Keep the latest callback in a ref (useLayoutEffect) so run() always calls it; then remove the argument workarounds. | Needs Investigation |
| QA-07 | Tab pills are 31 px tall on touch screens | Low | P3 | Shared primitives (PillSelector, TrainingTabs) | Open /workouts at 375 px; measure the today / calendar / history / templates pills. | Touch targets around 44 px (platform guidance; AA minimum is 24 px). | 31 px tall — passes WCAG 2.2 AA 2.5.8 but below the comfortable size. | DOM measurement at 375 px | min-h-11 on PillSelector buttons at small widths. | Open |
| QA-08 | Frontend and backend exercise datasets differ | Low | P3 | Exercise data | Compare frontend/src/lib/repdb.catalog.json (601) with the backend library after import (702). | One dataset version shared by both. | Different sizes; 13 curated backend rows use their own names. Name-based matching needed aliases and an externalId lookup. | db:import-repdb output; resolver fix FX-05 | Store RepDB ids (not names) in templates and sessions; regenerate the client catalog from the same dataset. | Open |
| QA-09 | Body-map screen: nested interactive controls and focusable hidden content | Medium | P2 | Body screen (/body), body-muscles library | 1. Open /body. 2. Press Tab repeatedly. 3. Run axe. | Each muscle reachable once with a clear name; no focusable content inside aria-hidden. | axe: nested-interactive and aria-hidden-focus (serious). The library renders every muscle path as role=button tabindex=0, so the chart adds about 70 tab stops. | docs/qa/evidence/axe-app-wide.json | Make the chart one composite control (roving tabindex), or make it inert and offer a muscle list as the accessible alternative. | Open |

## 3b. Defects fixed within the workout scope

| ID | Title | Sev. | Module | What was wrong | Fix | Verified by |
|---|---|---|---|---|---|---|
| FX-01 | Builder always saved the blank first-render draft | High | Workout builder | Save used a stale closure: API mode got a 400 'name required'; local mode created an empty nameless template. | Pass the draft, edit id and units as arguments to the save action. | E2E: create from scratch, edit and delete (API mode) |
| FX-02 | Editing a saved workout stuck on 'loading…' (dev) | Medium | Workout builder | A 'load started' ref skipped the StrictMode re-run while the first run's result was discarded. | Plain cancel-and-reload effect. | E2E: edit and delete a saved workout |
| FX-03 | 'Add a warm-up set' turned every set into a warm-up | High | Session start | All sets were flagged warm-up, so the exercise vanished from volume and PRs. | One warm-up set before the working sets (local and API starts). | E2E: 4 sets, exactly 1 warm-up, first |
| FX-04 | API mode lost RPE / warm-up (and timed / per-side) | Medium | Template persistence | WorkoutExercise has no columns for them. | Readable suffix in notes, e.g. '[rpe 8 · per side]', parsed back on load and on session start. | Browser: saved, reloaded, '2 × 30–45 s / side' intact |
| FX-05 | Six common exercises saved as a different variant | High | Template persistence | Name search plus 'first fuzzy hit' fallback; backend curated names differ for 6 RepDB ids. | Resolve through the RepDB id (backend /library/exercises/:slug also matches externalId); no fuzzy first hit. | All 95 template exercises resolve via the id path |
| FX-06 | Starting a workout silently discarded the open one | High | Workouts | Every start replaced the active session; the API session was orphaned. | Ask: resume / discard and start / cancel; discard abandons the API session. | E2E: start while open |
| FX-07 | Builder 'back' left the app when opened from a link | Low | Workout builder | history.back() with no in-app history. | Go to Workouts when there is no in-app history. | Browser |

| ID | Title | Sev. | Module | What was wrong | Fix | Verified by |
|---|---|---|---|---|---|---|
| FX-08 | Quick-action button covered the builder footer (mobile) | Medium | Layout | Floating + sat over 'cancel / discard changes'. | Hidden on the builder, like the active workout. | E2E @mobile builder; browser 375 px |
| FX-09 | Builder steppers clipped at phone width | Medium | Workout builder | Three steppers in a row overflowed; 3-digit values cut off. | 2×2 grid on phones; compact stepper text size. | E2E @mobile; browser 375 px |
| FX-10 | Stepper inputs had no accessible name | Medium | Shared Stepper | Visible label not associated with the input. | aria-label from the label (and unit). | E2E locates inputs by name |
| FX-11 | Filter selects announced every option as their name | Low | Template library | <label> wrapped the <select>. | htmlFor / id labels. | E2E getByLabel exact |
| FX-12 | Dark thumbnail placeholder looked like a hole | Low | Exercise media | Images sat on #1a0f16 until loaded. | Light placeholder matching the illustrations. | DOM check |
| FX-13 | 'Overhead Press' alias pointed at the barbell press | Low | Exercise matching | Regression from the Milestone 3 alias change; the backend row is the dumbbell press. | Alias restored to dumbbell-shoulder-press. | Resolver check |
| FX-14 | Finish screen and dialog backdrop clipped on the active workout | Medium | Active workout | The fixed overlays sat inside the animated route container (a transformed ancestor), so they were sized to it and cut off. | Render the finish screen and backdrop into document.body (portal). | Evidence screenshot desktop-10; session E2E |
| FX-15 | Finish-screen body map added ~140 tab stops | Medium | Active workout | The decorative front / back body map inherited the library's focusable paths inside aria-hidden. | The decorative map is inert. | axe scan of the finish screen in session E2E |
| FX-16 | Thumbnail motion missed a hover that began before the image loaded | Low | Exercise media | Hover listeners attached after load, so a pointer already over the card never started the motion. | Start immediately if the row is already hovered or focused. | Library E2E (motion allowed and reduced) |

# 4. UX / UI findings

| # | Finding | Heuristic / standard | Severity | Status |
|---|---|---|---|---|
| U-01 | Templates were a flat list with two chip rows; no search, no multi-day plans, no frequency or equipment information. | Recognition over recall; match to real world | High | Implemented |
| U-02 | Card secondary actions were small plain-text links; nothing signalled the card opened details. | Perceived affordance | Medium | Implemented |
| U-03 | Builder reorder / duplicate / remove icons were 14 px with no padding. | WCAG 2.5.8 target size; Fitts | Medium | Implemented |
| U-04 | No protection for unsaved builder changes; discard had no confirmation. | Error prevention | High | Implemented |
| U-05 | Save was simply disabled with no reason; no inline validation. | Visibility of status; help users recover | Medium | Implemented |
| U-06 | Every new exercise got 3 × 8–12 / 90 s, including stretches and cardio. | Sensible defaults | Medium | Implemented |
| U-07 | No way to replace an exercise; swaps had to be remove + add + re-enter values. | Efficiency of use | Medium | Implemented |
| U-08 | Finish screen lacked a payoff (no week progress, no visual of muscles worked, no celebration or sharing). | Feedback; motivation (cf. Hevy, Strong) | Medium | Implemented |
| U-09 | Exercise thumbnails were small, static and on a dark placeholder. | Aesthetic and minimalist design | Low | Implemented |
| U-09b | Exercise images: free-exercise-db photos have unknown provenance (maintainer says so in issue #2) and wger is per-entry CC-BY-SA. Kept RepDB and improved presentation: start ↔ peak crossfade, larger figures, light placeholder. | Licensing; recognition | Low | Implemented; animated tier is a purchase decision |
| U-10 | Tertiary text contrast below AA app-wide (QA-01). | WCAG 1.4.3 | Medium | Open |
| U-11 | Terminology mixes 'template', 'workout' and 'plan' (tab 'Templates', section 'my workouts', button 'save workout'). | Consistency and standards | Low | Open — rename the tab 'Library' |
| U-12 | On phones the Workouts header, training tabs and page tabs push content ~450 px down. | Progressive disclosure | Low | Open |
| U-13 | Motion respects prefers-reduced-motion: thumbnails and confetti stay still for those users (intended). | WCAG 2.3.3 | Low | By design |

## Accessibility scan (axe-core, WCAG 2.2 A/AA rules)

Workout screens (templates tab, builder, plan details dialog, finish screen) are asserted in E2E with no serious or critical violations, with color-contrast excluded because it is the app-wide token tracked as QA-01. App-wide report-only scan of 10 routes: color-contrast on 7 routes (/exercise-library 230 nodes, /workouts 22, /body 11, /settings 4, /progress 3, /dashboard 2, /nutrition 2); /body additionally fails nested-interactive and aria-hidden-focus (QA-09); /trainer, /store and /goals are clean. Raw results: docs/qa/evidence/axe-app-wide.json.

# 5. Technical findings

| # | Finding | Area | Severity | Status |
|---|---|---|---|---|
| T-00 | The chat E2E test accepts 200, 500 or 503 and checks the body only on 200, while success is 201, so it can never fail; it hid QA-10. | Testing | Medium | Open |
| T-01 | useAction memoises the first render's callback (QA-06); the builder save bug (FX-01) was this. | Data layer | Low | Needs Investigation |
| T-02 | Templates and sessions reference exercises by name; ids would remove the alias / fuzzy-match layer (QA-08). | Data model | Low | Open |
| T-03 | RPE / warm-up / timed / per-side are carried in WorkoutExercise.notes as a suffix; proper columns need an additive migration (approval). | Schema | Low | Open |
| T-04 | No frontend unit-test runner; logic was validated with scripts and E2E only. | Tooling | Low | Open |
| T-05 | React Compiler readiness warnings rose from 45 to 57 (new components use effect-driven state and random confetti positions). | Lint | Low | Open (warnings only) |
| T-06 | Dependency advisories (QA-02). | Supply chain | High | Open |
| T-07 | E2E ports are now configurable (E2E_WEB_PORT / E2E_API_PORT) and the stack script accepts a named disposable DB host; chat tests never bill a real key. | Testing | Low | Implemented |

# 6. Prioritized remediation backlog

## Now (before any public deployment)

| Item | Covers | Impact | Depends on | Size |
|---|---|---|---|---|
| Add the missing KaiMemory migration; make memory recall fail soft; tighten the chat test | QA-10, T-00 | Coach chat works on a deployed database | Migration approval | S |
| Upgrade vulnerable dependencies and re-run all suites | QA-02, T-06 | Security | Database for API tests | M |
| Fix the tertiary text token (#776E7C) and re-run axe | QA-01, U-10 | Readability for everyone; AA compliance | — | S |
| Deploy the API with a paid database | Limitations | Makes API-mode features real for users | Hosting decision (T-22) | M |

## Next

| Item | Covers | Impact | Depends on | Size |
|---|---|---|---|---|
| User-owned custom exercises (endpoint + schema) | QA-03 | No lost exercises in saved workouts | Migration approval | M |
| Self-host exercise images, or buy RepDB Standard for animations + heatmaps | QA-04 | Reliability; richer media | Purchase decision | M |
| useAction keeps the latest callback (ref) | QA-06, T-01 | Removes a class of stale-state bugs | — | S |
| Exercise ids instead of names in templates / sessions | QA-08, T-02 | Removes alias layer | Migration approval | L |

## Later

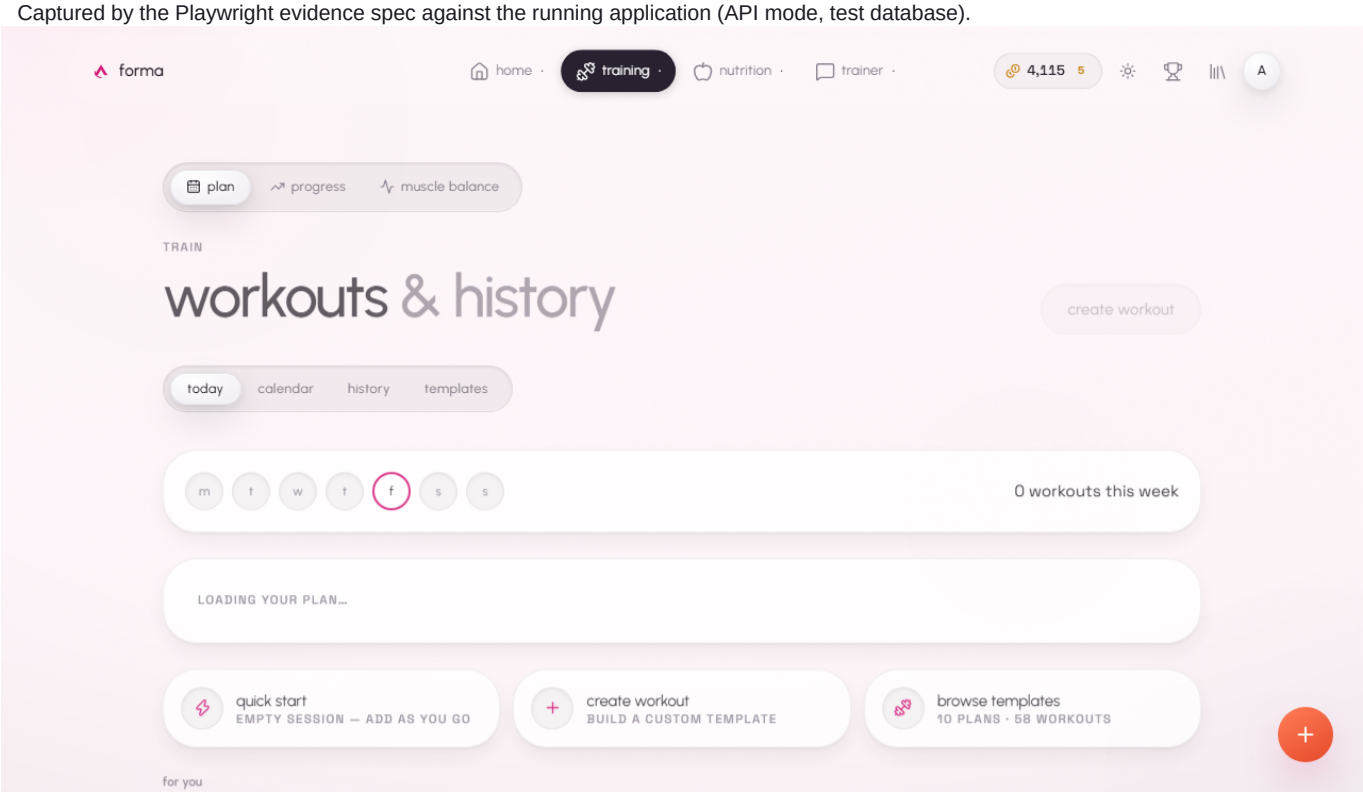
| Item | Covers | Impact | Depends on | Size |
|---|---|---|---|---|
| 44 px tab pills on touch screens | QA-07 | Comfort on phones | — | S |
| Rename 'Templates' tab to 'Library'; one term per concept | U-11 | Clarity | — | S |
| Stop signed-out polling | QA-05 | Cleaner logs | — | S |
| Frontend unit-test runner (Vitest) | T-04 | Faster feedback on logic | — | S |

# 7. Testing limitations

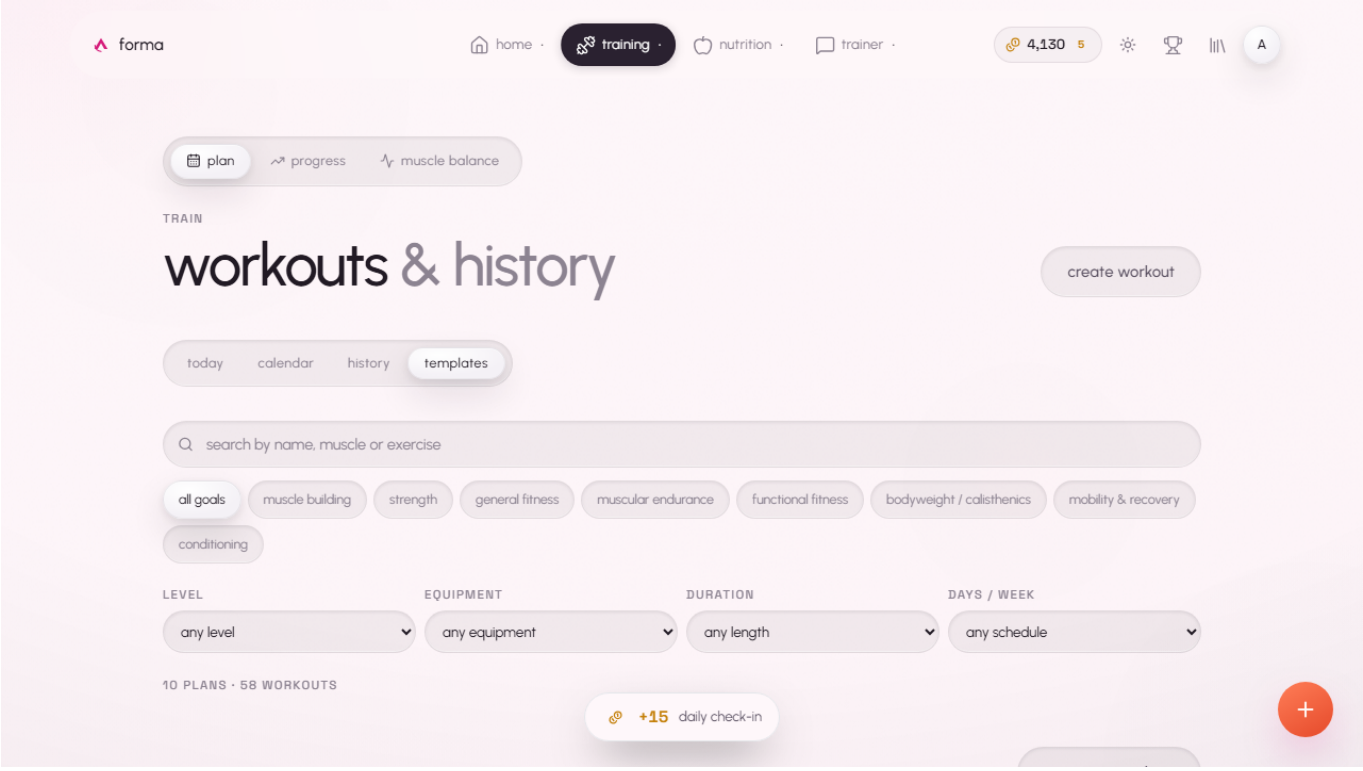
- Only Chromium was automated (desktop profile and an emulated Pixel 7). Safari / WebKit, Firefox and real devices were not tested.
- Accessibility was checked with axe-core and accessible-name assertions; no manual screen-reader pass (VoiceOver / NVDA) was done.
- The no-backend (GitHub Pages) build was not re-tested end to end in this pass; Playwright ran in API mode.
- Live Kai (Claude) replies were not exercised — the API key was cleared for tests to avoid billing.
- Payments / subscriptions, wearables OAuth and email delivery were not tested (no accounts or SMTP configured).
- The database is a remote disposable Render instance; latency is higher than production and it expires after 30 days.
- GitHub Actions CI has not run yet; results here are from local runs.
- The in-app browser reported prefers-reduced-motion, so thumbnail motion was verified only through Playwright's no-preference emulation.
- Performance (load times, bundle budgets) was not measured beyond the existing build size warning for the lazy exercise catalog.

# Appendix A — Screenshot evidence

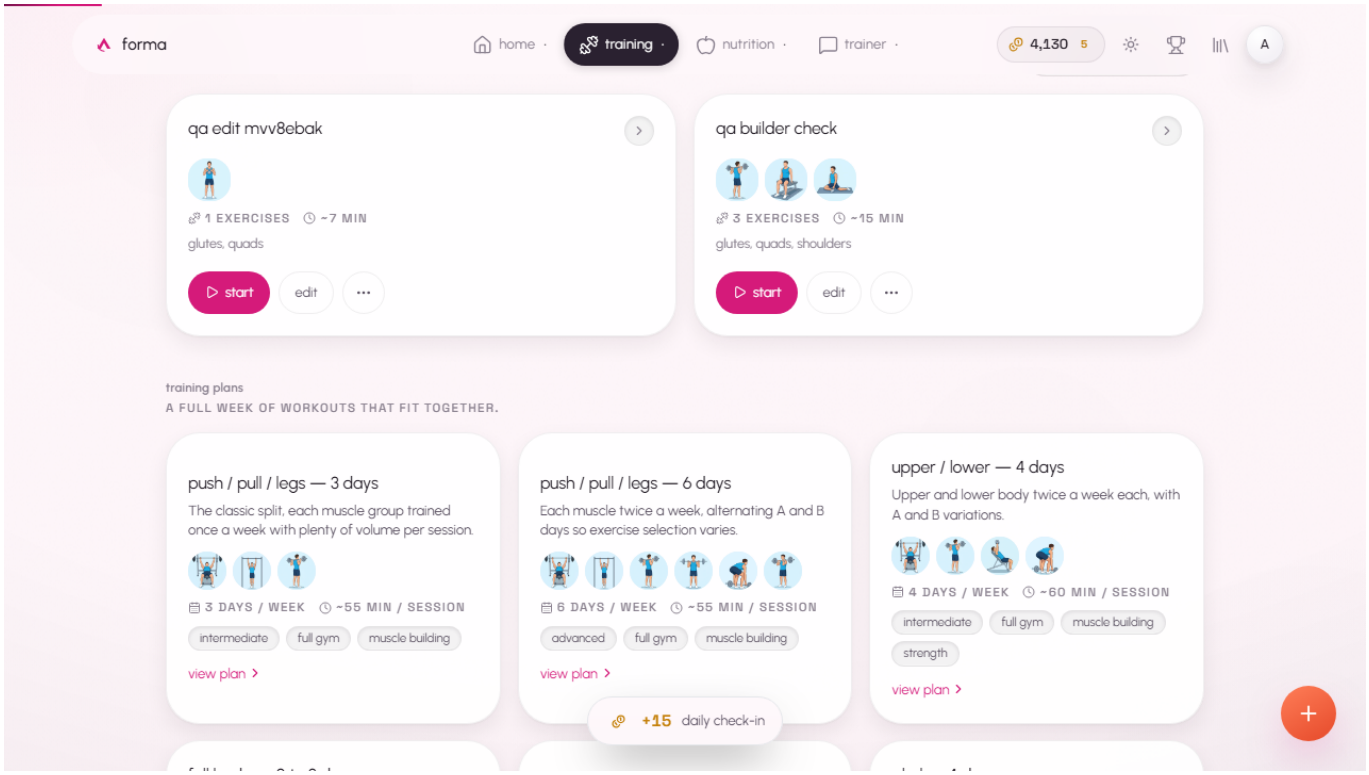
Captured by the Playwright evidence spec against the running application (API mode, test database).



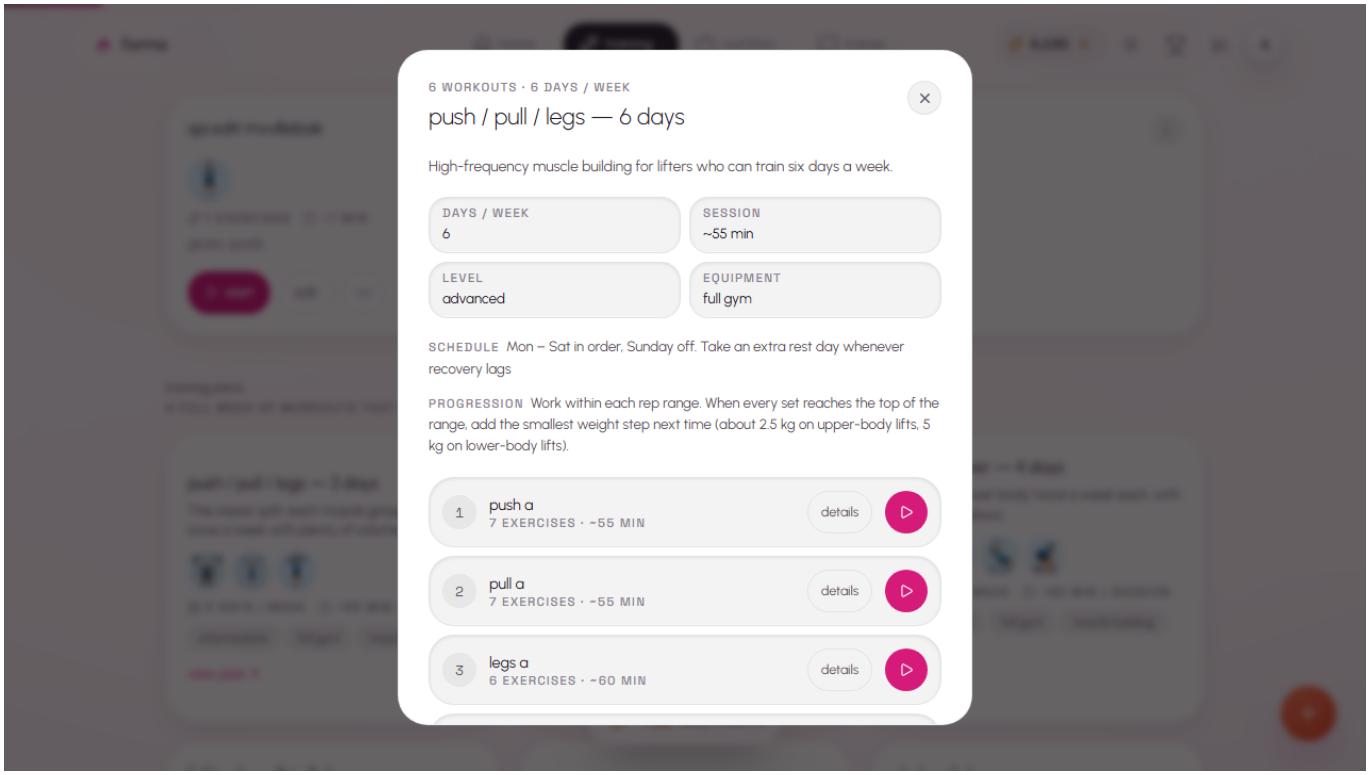

**S-01** — Workouts → Today with the new week strip



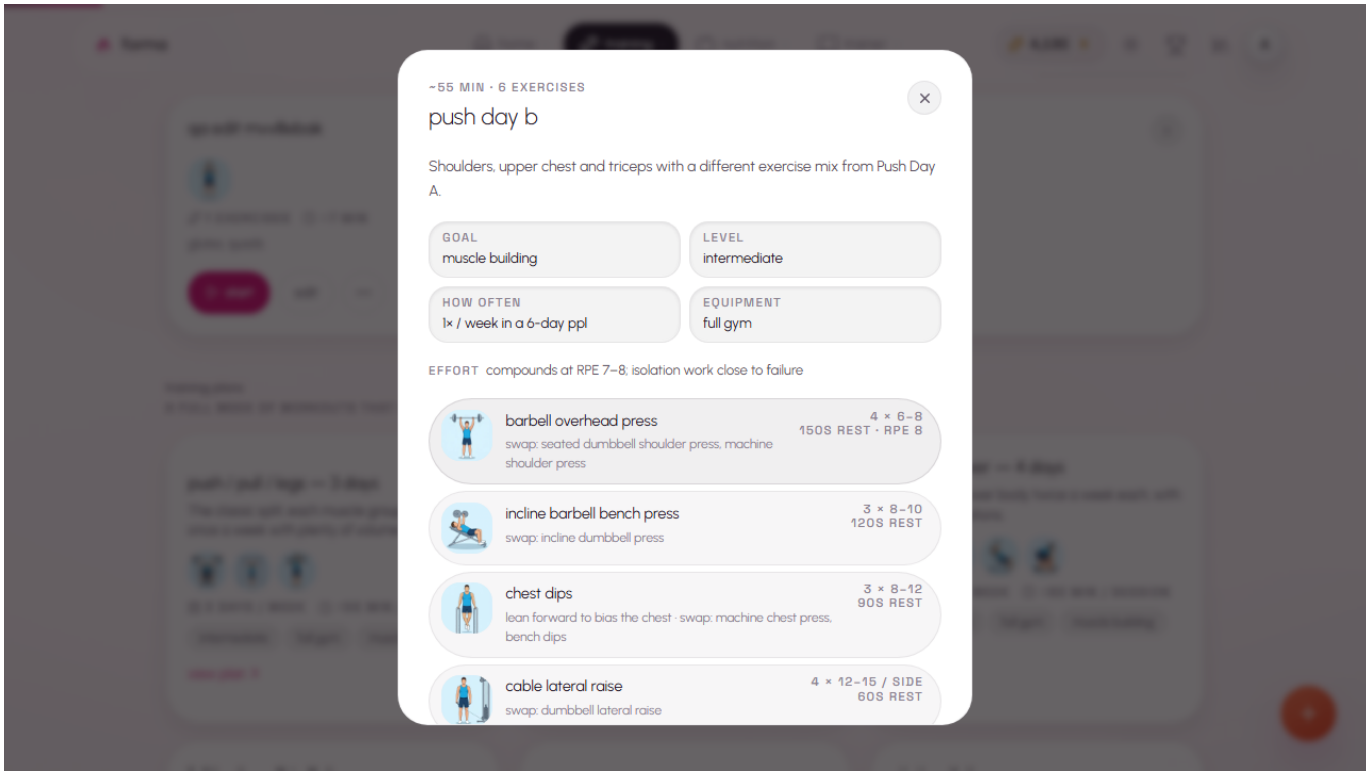

**S-02** — Template library: search, goal chips, filters, live count



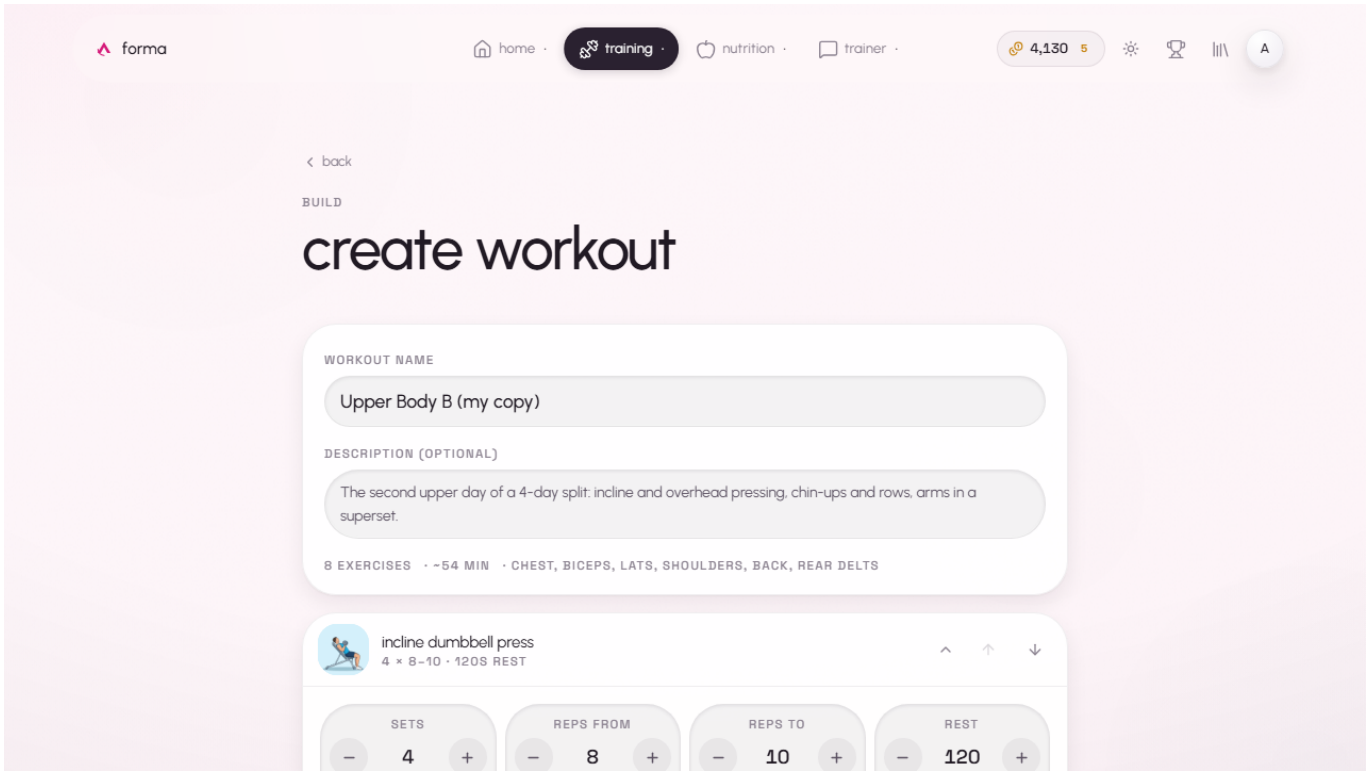

**S-03** — Training plans with day thumbnails, frequency, session length and tags



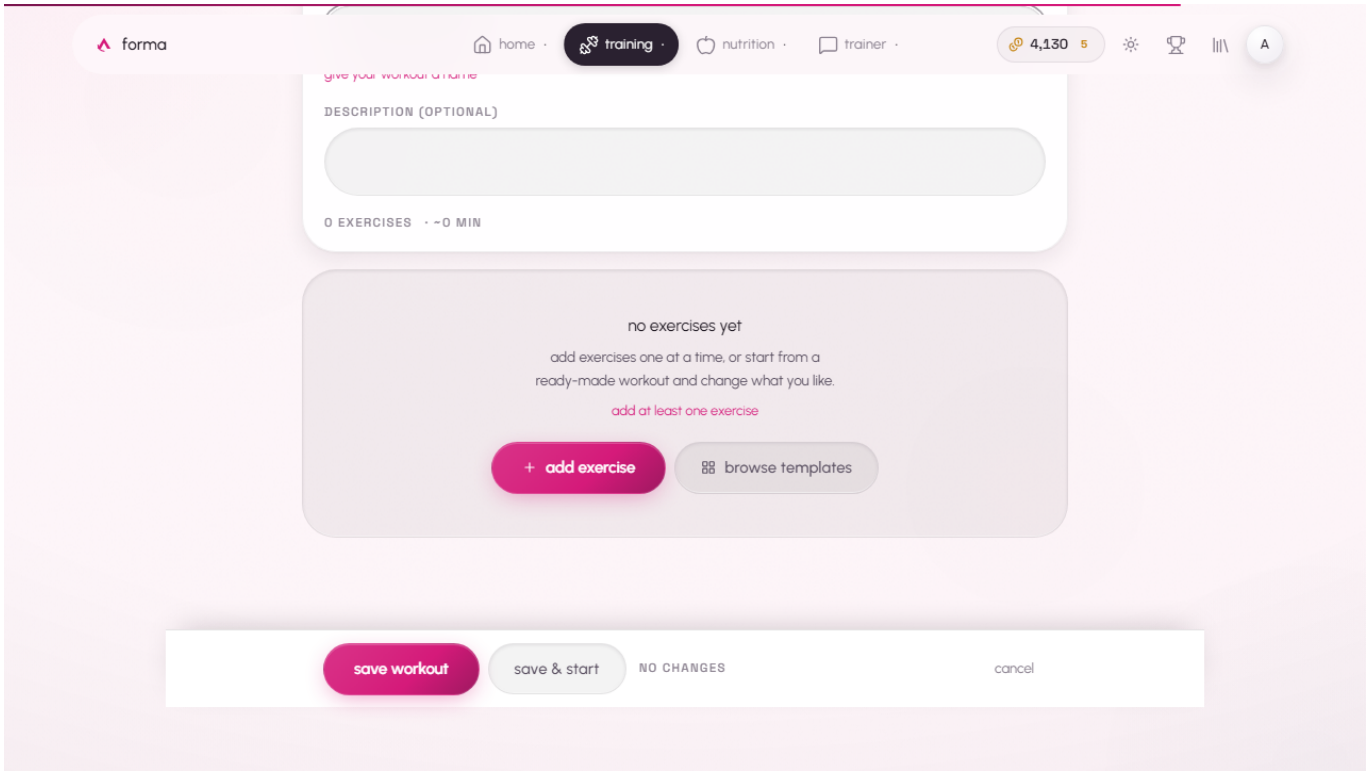

**S-04** — Plan details: schedule, progression and every day



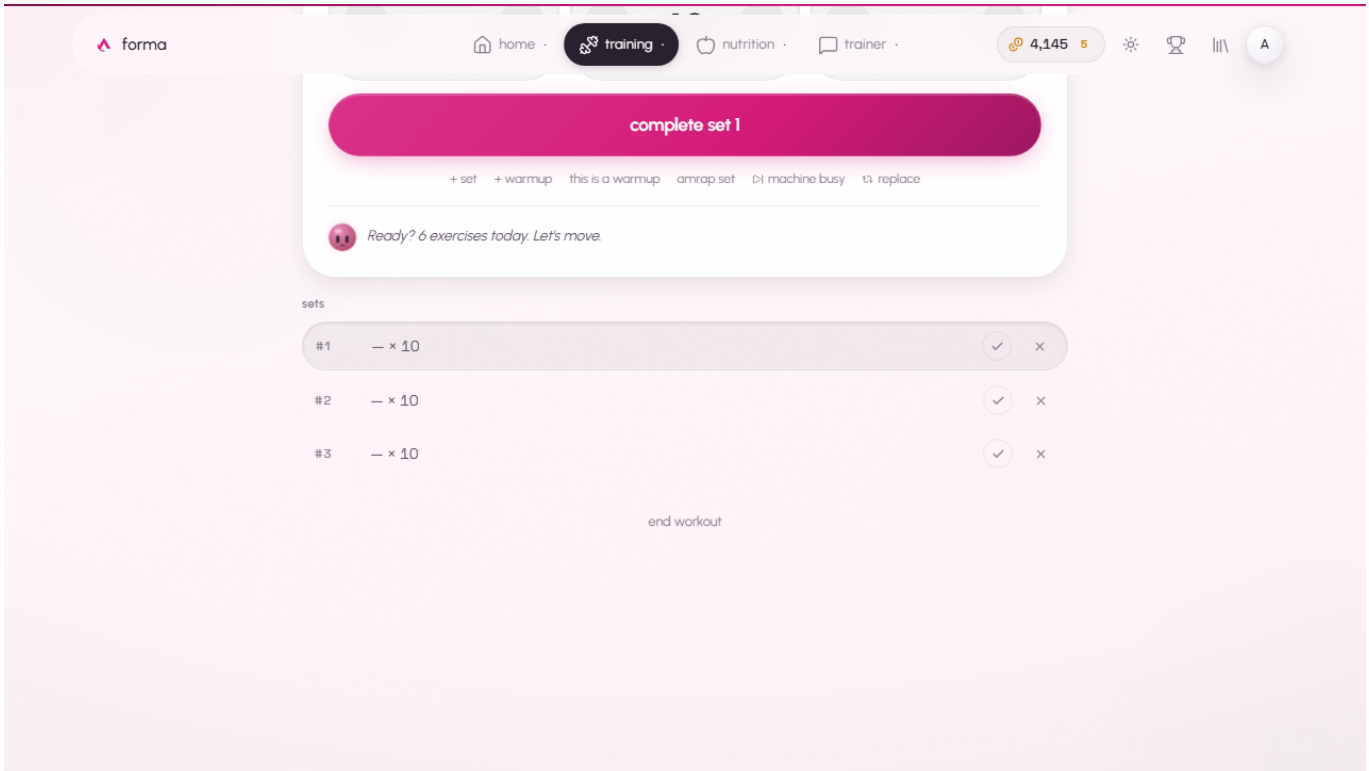

**S-05** — Workout details opened from a plan day



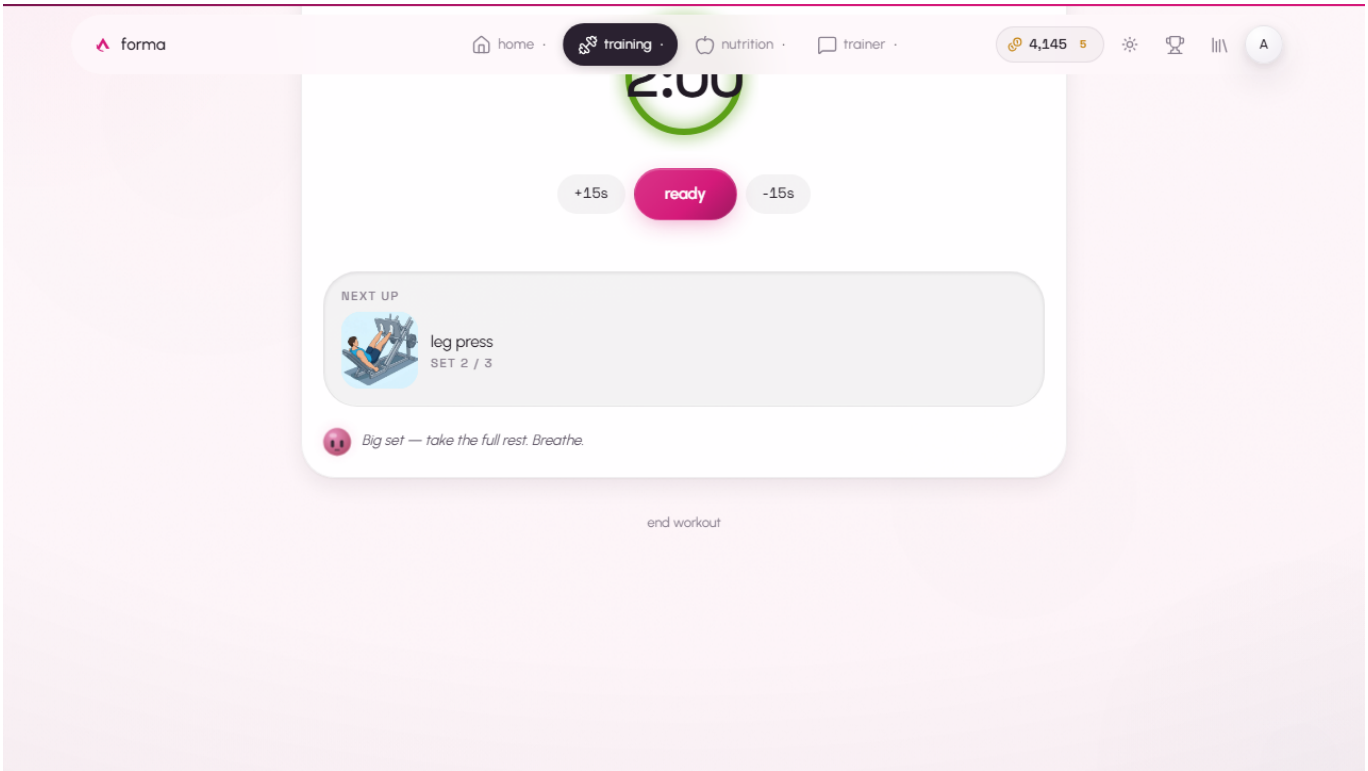

**S-06** — Builder: editing an exercise from a duplicated template



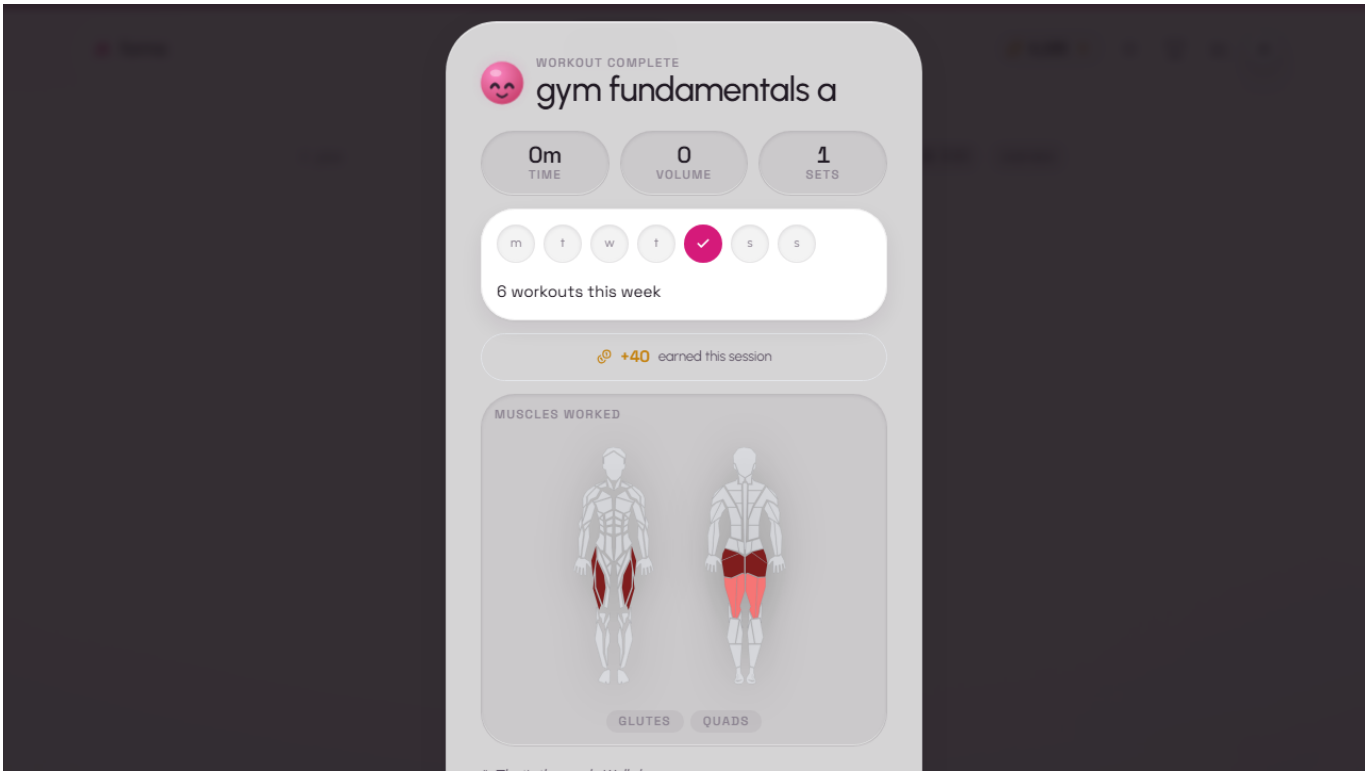

**S-07** — Builder: inline validation when saving an empty workout



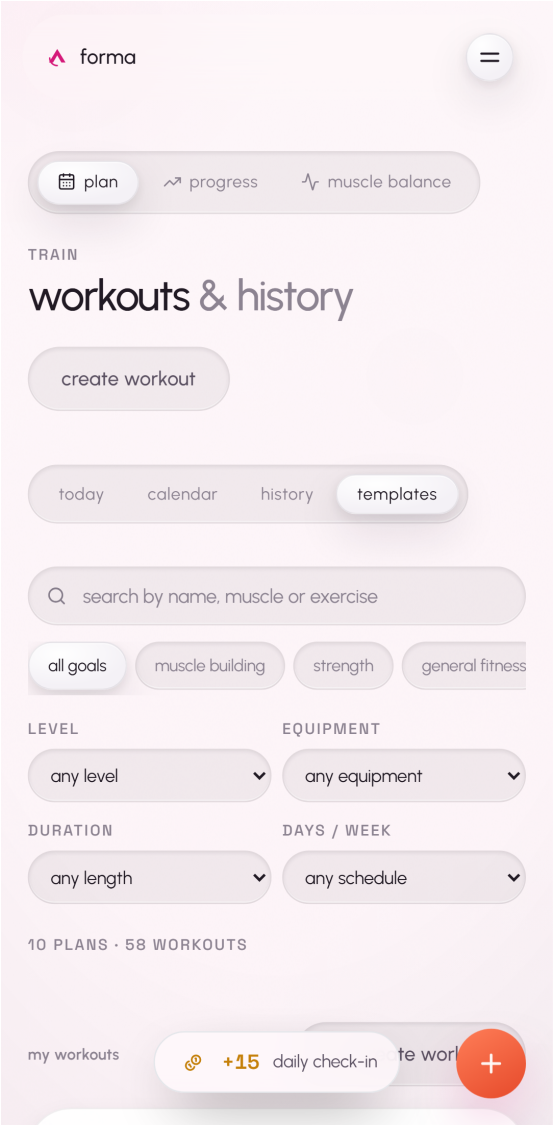

**S-08** — Active workout started from a template



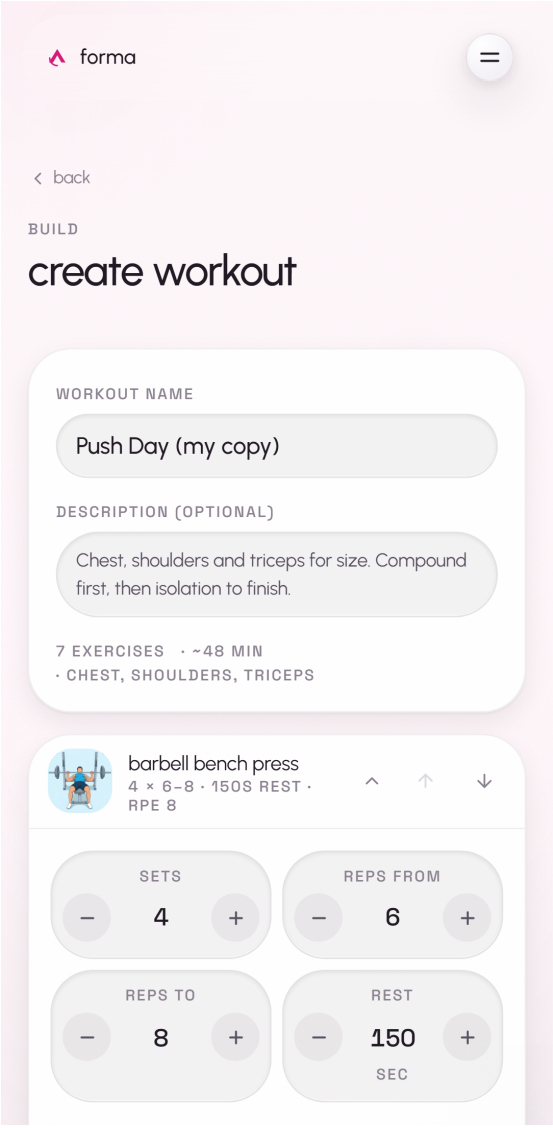

**S-09** — Rest timer with the larger looping next-up figure




**S-10** — Finish screen: week progress, muscles worked, coins




**S-11** — Phone (Pixel 7): template library




**S-12** — Phone (Pixel 7): builder steppers in a 2 × 2 grid

# Appendix B — Changed files (this pass)

| File | Change |
|---|---|
| frontend/src/lib/workoutTemplates.ts | 20 new templates; goals, frequency, timed / per-side flags; filter helpers; notes-suffix encoding; warm-up sets |
| frontend/src/lib/workoutPlans.ts (new) | 10 multi-day training plans and plan helpers |
| frontend/src/lib/templates.ts | Resolve exercises through the RepDB id; unmatched-exercise check; encode flags in notes |
| frontend/src/lib/lifecycle.ts | API session start decodes flags; one warm-up set |
| frontend/src/lib/repdb.ts | Start / peak frame lookup; alias fixes; exported name helpers |
| frontend/src/lib/fitness.ts | weekProgress for the week strip |
| frontend/src/components/ExerciseThumb.tsx | Name-based frames, start ↔ peak crossfade, light placeholder, reduced-motion aware |
| frontend/src/components/workout/TemplateLibrary.tsx (new) | Search, goal chips, filters, live count, sections |
| frontend/src/components/workout/TemplatePreview.tsx (new) | Workout details drawer |
| frontend/src/components/workout/PlanViews.tsx (new) | Plan card and plan details drawer |
| frontend/src/components/workout/WeekStrip.tsx (new) | This-week progress and streak |
| frontend/src/components/workout/Celebration.tsx (new) | Finish-screen confetti (off with reduced motion) |
| frontend/src/components/workout/TemplateCard.tsx | Action menu, meta chips, exercise thumbnails, 40 px targets |
| frontend/src/components/workout/ExercisePicker.tsx | Suggestions, match count, all-muscles chip, pressed states |
| frontend/src/components/workout/Stepper.tsx | Accessible name; compact size |
| frontend/src/components/workout/SessionDetailDrawer.tsx | Name-based thumbnails |

| File | Change |
|---|---|
| frontend/src/components/layout/QuickActions.tsx | Hidden on the builder |
| frontend/src/pages/Workouts.tsx | Library, detail drawers, save plan, rename, start-while-active guard, week strip |
| frontend/src/pages/WorkoutBuilder.tsx | Rewritten: smart defaults, validation, replace / duplicate / undo, unsaved-changes guard, draft restore |
| frontend/src/pages/ActiveWorkout.tsx | Finish screen payoff (week, muscles worked, PR headline, share); portal overlays; larger next-up figure |
| frontend/src/pages/ExerciseLibrary.tsx | Name-based thumbnails |
| backend/src/modules/library.ts | GET /library/exercises/:slug also matches the RepDB externalId |
| frontend/tests/e2e/*.spec.ts (6 new) | workout-library, workout-builder, workout-session, smoke, export, evidence |
| frontend/playwright.config.ts, tests/e2e/helpers.ts, scripts/e2e-stack.mjs | Configurable ports, Pixel 7 project, disposable DB host |
| frontend/package.json | @axe-core/playwright |